

Film Digitization Research Proposal

Megan Jordan

Department of Library and Information Sciences, University of Denver

RMS 4900: Education and Research Measurement

Dr. Spencer Acadia

June 5th, 2025

1. Topic Under Investigation & Purpose of Study

The topic of this proposed research study is workflows and standards in film digitization. Film digitization is a complicated, time consuming, and often nuanced process. Due to rapidly changing technology, standards for this process have never been truly formulated. This leaves a huge gap in literature and resources for film archives and memory institutions seeking a standard or looking for workflow guidance in how to set up their digitization projects and to keep their repositories up to date. Research is needed on this topic to help these digitizers understand common current practices with which to begin formulating a foundation to build a standard and workflow processes that will support and enlighten the film digitization field.

This study is a qualitative study to understand how film archives and film repositories are currently digitizing their film. It will include every stage of digitization and will use multiple data collection methods to ensure that the richest and most thorough data is collected on this topic. The aim of the study is to determine what the current film digitization and workflow practices are at film archives, film repositories and memory institutions. Additionally, another aim is for the data to be analyzed to understand if it can be used to build a standard for film digitization.

The benefit of this study is to current and future film digitization archives, memory institutions and repositories. Many institutions and digitizers struggle to determine use neutral standards for their film digitization projects. Creating comprehensive and easy to access data on current digitization practices will benefit digitizers who are looking to update their standards or workflows and to create solid, future proof, digitization projects. While the study itself will not create a standard, the analyzed data will be integral in creating an understanding of common practices and how they align with a standard that might be built from the collected data.

The value of the study is in finding actual, real-world data to support digitizers who struggle to determine their own project standards and locate case studies to align their own digitization projects to. This data would save many digitizers time and effort in researching and determining these standards on their own without a baseline to start from. It is also valuable to create use neutral digital objects so that the digitized film can continue to be accessed and preserved through future technology changes, ensuring the original film can be preserved, or the film will live on in the digital form if the original film is lost.

2. Theoretical or Philosophical Perspective

While no standard exists for film digitization, the Federal Agencies Digitization Guidelines Initiative (FADGI) has spearheaded creating a document that outlines best practices for digitizing film, titled *Digitizing Motion Picture Film* (2016). The 23-page guide outlines basic concepts and guidelines for film digitization. It covers basic definitions for film digitization institutions and outlines the basic process including inputs, scanners, transfers, encoding and outputs. It also offers guidance for institutions looking to outsource their digitizing to ensure they are armed with enough information to receive the type of digitized files they are seeking.

However useful the guidebook is, it is very limited in scope and even self-proclaims that it uses an “easy” use case for its only main sample case and guide. Regardless of this limitation, the FADGI guide is a great start for a niche that requires a much deeper investigation into its many complexities and the overview is helpful for beginners as an introduction into film digitization. The guide can also be used as additional data points within the study. It will be useful to use this guide as a comparison to the data collected from this study to understand if the standards proposed in the guide are common practice for film digitizers. This would help build a

case for determining is what is in the guide can be used as a standard, or also to propose revisions to the guide for a different standard.

3. Research Question(s)

The proposed research question is: How are film archives, memory institutions and film repositories digitizing their moving image collections? This question encompasses both the tools these institutions are using, such as hardware and software, as well as workflow processes.

It is important to define moving images to understand the scope of this study. Moving image is a term that was coined in the early days of film, referring to photos that appeared to move when viewed in quick succession (which is exactly what they are, more commonly known as movies). Moving images are a timecoded analog medium. Cellulose film is the physical medium on which many moving images are recorded, referred to as a film negative. For the purposes of this research proposal, the scope of research will be limited to digitization related to this type of film negative, excluding other, newer types of moving image mediums such as tape (VHS, MiniDV, etc) and digital born video. This will include various film stock sizes, including 8mm, 16mm, 35mm and 70mm. Because of this limitation and specification, the terms film and moving image will be used interchangeably in this proposal. Films do include more commonly known entertainment such as movies but can also include home movies, oral histories, documentary footage and more.

Film digitization is also important to define, which means creating a digital surrogate of an original analog film. These surrogates will theoretically help institutions create access points for future users and researchers, or more likely due to copyright, a searchable finding with rich metadata for a researcher to find and inquire about. The digitization process in many cases will also allow the original film to stay preserved and safely stored and unused. All film eventually

degrades and becomes unusable, and the less film is touched and projected, the longer its life can last. In this way, digitization has benefits for both access and preservation.

The final term requiring definition is use neutral. This term is used across digitization efforts in library and information science and is not specific to film. It refers to digital objects being saved and stored in ways that are open, nonproprietary and easily transferable. Essentially, future proofing files so items do not need to be redigitized as technology inevitably changes or after a film is no longer useable. While this term is not integral to the methodology, it has already been referenced and is an important part of film digitization workflows and end point access.

4. Methodology

This study will be qualitative and be completed in two parts. Digitizing film is not a simple or even straightforward process. There are many steps and a number of different softwares, hardwares, processes and tools that need to be utilized, and many options for digitizers to choose from within each step of the process. Using a qualitative method will enable the researcher to collect rich and nuanced data that quantitative data would miss. Due to film digitization not having a standard and the many commercial options on the market for each step, there are a vast number of answers that a Likert scale or multiple-choice survey could not include or that the researcher may not know to include. Qualitative methods will allow for the most accurate answers as well as provide the opportunity for in depth and nuanced answers about digitization processes and how digitizers feel their processes are working or could be improved upon.

The two-part study will be done sequentially. The first part will be a qualitative survey. A set of open-ended questions, designed to take 20 minutes or less to answer, will be created as a survey in JotForm. The survey will include straight forward open-ended questions about tools

and process, as well as more thought-provoking questions about challenges and opportunities in the institutions digitizing process. The surveys will also include questions about basic institutional data such type, location and collection size. Once those surveys are returned, I will code, review and analyze the results. With those results, I will formulate a set of interview questions and interview 5-10 additional digitizers to interview over phone or video conference. These questions are intended to give the first set of answers more context and to clarify any questions that may have arisen out of the survey questions. These interview responses will also be coded, reviewed, analyzed, and integrated into the survey results.

The data needed to answer my research question is the type of software, hardware, tools, workflows and systems digitizers are using to digitize film. This would also include a breakdown of the steps of their workflows. Some institutions may outsource their digitizing while others may have in-house systems. Other institutions may have one person who is responsible for digitization efforts while others may have a team of people responsible. The workflow that the institution uses and the systems they have chosen to digitize their film are the key components to the data required.

The data will be collected via a JotForm survey for the first part of the study and by recording for the second set of interviews. A post with an explanation of the research study and an email to contact will be posted on ListSrvs, forums, and sent to emails associated with film archives, repositories and memory institutions. This convenience sampling method will be used to get the broadest sampling possible with the limited resources of the study. This is the simplest way to find the largest audience of potential participants.

ListSrvs associated for film archivists such as AMIA (Association of Moving Image Archivists), SAA (Society of American Archivists), etc will be targeted. I will also research as

many institutions as possible that have digital film archives and reach out directly to staff who would be involved in digitization efforts. Those staff would have titles such as Preservation Specialist, Digital Archivist, Media Digitization Manager. For an institution without a directory of employees, I will email the Archives or Collections and Preservation general email.

The posting on the ListServes will ask the recipients to participate in the initial survey. A positive response to the designated email will allow me to email them the link to the survey. After the first round of surveys are received, a second posting will be posting and more emails directly to institutions will be sent, including additional institutions that did not participate in the first round of survey. The initial posting may have been missed by certain participants who would have been interested, and the second posting will give them an opportunity to participate in the second part of the study. It will also give those who saw the first posting but may not have had bandwidth to participate in the first part of the study but now have the time to do so.

5. Ethical Considerations

When participants volunteer, they will be required to sign an Informed Consent form before the JotForm survey will start. A thorough explanation of the study and its purpose, as well as withdrawal instructions, will be supplied to all potential participants. The explanation will also confirm to participants and digital repositories that they will have their data kept confidential.

Interviewees will have their names and/or institutions coded with their notes, so their identity remains confidential. Once the data has been coded, the digital record of the surveys and the key to the codes, will be downloaded onto a password protected hard drive and deleted from the JotForm website.

In addition, I will create a special Gmail account to use for correspondence for this study. This will ensure that any personal information and confidential study results are not mixed in my

personal email. When the study is over, this Gmail can also be moved to the password protected hard drive and its contents deleted.

The study is inquiring about digitization process, not examining digitized files themselves. This will alleviate some concerns that may have arisen if the films themselves were being examined, as that might induce been more participant stress depending on the nature of the films i.e. speaking frankly about dissatisfaction with institution decisions, copyright concerns or sharing personal home movies in a collection. That does not mean that the study will not have any stress or involve no vulnerable populations, but its goal is to make those who are willing to participate feel comfortable to speak freely about their processes.

This encapsulates this film digitization research proposal. The study will collect and analyze important data that may feel out of reach to many of those working in this field. Completing this study and making its results available to film digitizers at large is an important step toward creating a film digitization standard and supporting the film digitizing community.

6. References

Federal Agencies Digital Guidelines Initiative (FADGI). (2016, April 18). *Digitizing Motion Picture Film*. https://www.digitizationguidelines.gov/guidelines/FilmScan_PWS-SOW_20160418.pdf